

Quick Map

Attribute	Explanation
Structure	<!DOCTYPE html>, <html>, <head>, <meta>, <title>, <body>
Form	<form>, <fieldset>, <legend>, <label>, <input>, <textarea>, <button>
Table layout	<table>, <tr>, <td>
Important attributes	id, name, for, type, placeholder, required, checked, size, min, max, rows, cols, colspan, rowspan, cellpadding, cellspacing, align, style

1. <!DOCTYPE html>

Tells the browser that the page is an HTML5 document. It is written before the <html> tag.

Example

```
<!DOCTYPE html>
```

Output: No visible output. It tells the browser how to interpret the document.

2. <html>

The root element of an HTML document. All other HTML elements are written inside it.

Important attributes

Attribute	Explanation
lang	Specifies the language of the page. Example: lang="en" for English.

Example

```
<html lang="en"> ... </html>
```

Output: No direct visible output.

3. <head>

Contains page information such as metadata and the page title. Its contents are generally not shown in the page body.

Example

```
<head> ... </head>
```

Output: No direct visible output.

4. <meta>

Provides metadata about the page. In your solution it is used for UTF-8 character encoding and responsive viewport settings.

Important attributes

Attribute	Explanation
charset	Defines the character encoding, commonly UTF-8.
name	Names the metadata item, such as viewport.
content	Contains the value/settings of the metadata.

Example

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Output: The browser uses UTF-8 and the page can scale correctly on different screen sizes.

5. <title>

Sets the title displayed in the browser tab.

Example

```
<title>FORM 4</title>
```

Output: The browser tab shows FORM 4.

6. <body>

Contains the visible content of the webpage.

Example

```
<body> ... </body>
```

Output: The form, table and controls are displayed here.

7. <fieldset>

Groups related form controls inside a bordered box. It is useful when several controls belong to one section.

Important attributes

Attribute	Explanation
disabled	Can disable all controls inside the fieldset.
name	Gives the fieldset a name; less commonly needed.

Example

```
<fieldset>
  ... form controls ...
</fieldset>
```

Output: A bordered group appears around the controls.

8. <legend>

Provides a caption/title for a <fieldset>. It belongs inside the fieldset.

Important attributes

Attribute	Explanation
align	Legacy presentation attribute used in older examples. Modern pages should use CSS for alignment.

Example

```
<fieldset>
<legend align="center">CREATE ACCOUNT</legend>
</fieldset>
```

Output: CREATE ACCOUNT appears on the fieldset border.

9. <form>

Defines a form used to collect user input. The controls inside it can be submitted together.

Important attributes

Attribute	Explanation
action	Specifies where the submitted form data should be sent.
method	Specifies how data is sent, commonly GET or POST.
style	Applies inline CSS to the form.

Example

```
<form action="/submit" method="post">
...
</form>
```

Output: The form itself may not have a visible special appearance; its controls are shown.

10. <table>

Creates a table with rows and columns. In your Lab 4 solution, it is used to align labels and form controls.

Important attributes

Attribute	Explanation
border	Legacy way to add a table border.
cellpadding	Adds space INSIDE each cell, between the cell content and its border.
cellspacing	Adds space BETWEEN adjacent cells.
width	Sets table width; CSS is preferred today.
align	Old way to align the table; CSS is preferred.

Example

```
<table border="1" cellpadding="8" cellspacing="4" align="center">
...
</table>
```

Output: A table appears with internal cell padding and space between cells.

TABLE SPACING

CODE

```
<table border="1" cellpadding="8"
      cellspacing="4">
```

OUTPUT
cellpadding = space INSIDE cell
cellspacing = space BETWEEN cells

11. <tr>

Creates one row in a table. Each row normally contains <td> or <th> cells.

Example

```
<tr><td>Name</td><td>Yug</td></tr>
```

Output: One horizontal row is created.

12. <td>

Creates a normal data cell in a table. It can contain text, labels, inputs or other HTML elements.

Important attributes

Attribute	Explanation
colspan	Makes one cell span multiple columns.
rowspan	Makes one cell span multiple rows.
align	Legacy horizontal alignment attribute; CSS is preferred.
valign	Legacy vertical alignment attribute; CSS is preferred.

Example

```
<td colspan="2">Email*</td>
<td rowspan="2">Common</td>
```

Output: The first cell covers two columns; the second example cell covers two rows.

COLSPAN vs ROWSPAN

CODE

```
<td colspan="2">Email</td>  
<td rowspan="2">Common</td>
```

OUTPUT

colspan = merges/spans COLUMNS
rowspan = merges/spans ROWS

★ Table Attributes - Do Not Miss These

These are easy viva questions. Remember the direction: colspan → columns, rowspan → rows.

Attribute	Explanation
border	Adds a visible border around the table/cells. Legacy HTML presentation; CSS is preferred.
cellpadding	Space INSIDE a cell, between its content and border.
cellspacing	Space BETWEEN two adjacent cells.
colspan	Makes a cell cover multiple columns. Example: colspan="2".
rowspan	Makes a cell cover multiple rows. Example: rowspan="2".
width	Controls table width. CSS width is preferred.
align	Old alignment attribute. CSS text-align/margin is preferred.

Memory trick: ■ CELLPADDING = padding is inside. ↔ CELLSPACING = gap between cells.

↓ ROWSPAN = rows. → COLSPAN = columns.

13. <label>

Displays descriptive text for a form control. A label makes the form easier to understand and improves accessibility.

Important attributes

Attribute	Explanation
for	Connects the label to an element whose id has the same value.

Example

```
<label for="email">Email</label>  
<input type="email" id="email">
```

Output: Email appears beside the control. Clicking the label can focus/select its associated control.

LABEL + FOR + ID

CODE

```
<label for="email">Email</label>
<input type="email" id="email">
```

OUTPUT

for = points to id
Matching values connect label + input

14. <input>

Creates a form input control. The type attribute decides what kind of control is displayed.

Important attributes

Attribute	Explanation
type	Defines the input type.
id	Unique identifier for the element; used by label's for.
name	Name of the form field. Radio buttons with the same name form one group.
placeholder	Temporary hint shown inside an empty input.
required	Makes the field required for successful form submission.
checked	Preselects a radio button or checkbox.
size	Controls visible width for certain text-like inputs.
min	Minimum allowed value for suitable number/range inputs.
max	Maximum allowed value for suitable number/range inputs.

Example

```
<input type="text" id="firstName" name="firstName"
      placeholder="Enter your first name" required>
```

Output: A text box with a hint is shown. The field is required when the form is submitted.

Input Types Used in Lab 4

Attribute	Explanation
text	Single-line text such as first name or last name.
email	Email input with browser email-format validation.
password	Text is hidden while typing.
radio	Selects one option from a group. Use the same name for the group.
checkbox	Allows independent selections; multiple boxes can be selected.
number	For numeric values such as age. min/max can limit the value.
tel	The tel input type is used to create a field for entering a telephone/mobile number .
range	Displays a slider. min/max define its range.
file	Lets the user choose a file from the device.

Example:

```
<input type="radio" name="gender" id="male" checked>
<label for="male">Male</label>

<input type="range" min="0" max="20000" id="income">
```

for, id and name - Must Know

Attribute	Explanation
for	Used on <label>. It should match the target element's id.
id	Unique identifier of an element. Used by label, CSS and JavaScript.
name	Identifies the field when form data is submitted. Same name groups radio buttons.

Example:

```
<label for="male">Male</label>
<input type="radio" id="male" name="gender">
```

Here for="male" and id="male" match. The name="gender" groups the radio buttons.

FOR = ID

CODE
for="male" → id="male"

OUTPUT
The label is linked to the input.

15. <textarea>

Creates a multi-line text input. It is useful for longer information such as a Bio or Address.

Important attributes

Attribute	Explanation
rows	Approximate number of visible text lines.
cols	Approximate number of visible character columns.
id	Unique identifier, which can be connected to a label.
name	Name used when the textarea value is submitted.
placeholder	Temporary hint inside the empty textarea.
required	Makes the textarea required before successful submission.

Example

```
<textarea id="bio" rows="5" cols="30"
placeholder="Write your bio"></textarea>
```

Output: A multi-line text box appears.

16. <button>

Creates a clickable button. For a form, the usual button types are submit, reset and button.

Important attributes

Attribute	Explanation
type	submit submits the form; reset resets form controls; button creates a normal button.
style	Applies inline CSS.

Example

```
<button type="submit" style="background-color: teal; color: white;">CREATE</button>
```

Output: A CREATE button appears. With type="submit", it can submit the form.

Quick Viva Revision

Question	Short Answer
What is <fieldset>?	Groups related form controls inside a bordered box.
What is <legend>?	Gives a title/caption to a fieldset.
Is <legend> mandatory?	No, but it is recommended. When used, it belongs inside <fieldset>.
What is <label>?	Provides a name/description for a form control.
What is for?	Connects a label to an element using its id.
What is id?	A unique identifier for an HTML element.
What is name?	Identifies form data and groups radio buttons.
Default input type?	If type is omitted, the default is text.
cellpadding vs cellspacing?	cellpadding = inside the cell; cellspacing = between cells.
colspan vs rowspan?	colspan = columns; rowspan = rows.
What does required do?	Requires a value before successful form submission.
What does checked do?	Preselects a radio button or checkbox.
What does placeholder do?	Shows a temporary hint in an empty control.
What does min/max do?	Sets minimum/maximum for suitable number/range inputs.